

Study on the Influence of Defective Graphene on the Li Diffusion Performance in Si/Defective Graphene Composite Anodes: an Ab Initio Molecular Dynamics Study

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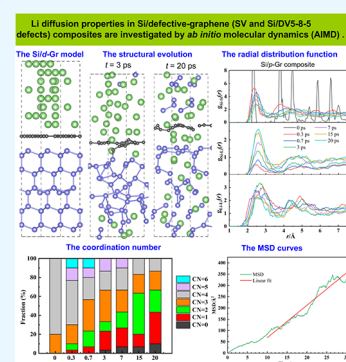
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ABSTRACT: Si/C composites are the most promising alternatives for anodes of lithium-ion batteries, as they can effectively address the challenges associated with volume expansion and structural degradation commonly encountered in Si-based anodes. Research results showed that the Li diffusion performance in Si/C composites was highly dependent on the C-layer structure, where ion channels were needed for Li diffusion. In order to illustrate how defective graphene promotes Li diffusion behaviors, the Li diffusion properties in Si/defective-graphene (SV and Si/DVS-8-5 defects) composites are investigated by *ab initio* molecular dynamics (AIMD) in this paper. The structural evolutions of Si/defective-graphene composites during the Li diffusion process are analyzed, and the effects of various graphene defects on Li diffusion are presented. The mean squared displacements of Li atoms are studied, and the Li diffusion coefficients are calculated. It is discovered that the defective-graphene layer could provide voids for Li atoms to diffuse in; therefore, the Li diffusion speeds in Si/defective-graphene composites are faster. The diffusion coefficients for Li atoms are increased from 1.88×10^{-4} to 2.47×10^{-4} cm²/s by the introduction of defects into Si/graphene composites.



1. INTRODUCTION

Lithium-ion batteries (LIBs) with high energy density, quick charging rates, and long cycle life are increasingly in demand with the development of various industries reliant on energy storage systems.^{1,2} Currently, graphite is the most commonly used anode material in commercial LIBs. However, graphite has a limited theoretical capacity of 372 mAh/g, which restricts the energy density of the battery. Hence, it is crucial to explore alternative anode materials that possess higher theoretical capacities in order to enhance the energy storage capability of LIBs.³

Silicon (Si) is one of the most promising candidates for next-generation anode materials due to its high theoretical capacity of about 4200 mAh/g.⁴ However, Si suffers from significant volume expansion during lithiation, leading to mechanical degradation and capacity fading over a repeated lithiation and delithiation process. Therefore, these challenges should be addressed in order to achieve the large-scale application of Si-based anode materials in LIBs.⁵

Two typical measures, namely Si nanostructuring^{6,7} and combining Si with carbonaceous materials (Si/C composites),^{8,9} are widely used to enhance the performance of Si anode materials in LIBs. A variety of Si nanostructured materials have been demonstrated to mitigate the mechanical stress caused by volume expansion and improve the rate capability of the battery, such as Si nanowires,¹⁰ porous Si particles, and Si nanotubes.¹¹ However, there are also some

drawbacks of Si nanostructured anode materials, for example, high production cost, easy agglomeration, low packing density, and increased side reactions.¹² Compared with Si nanostructured materials, Si/C composites have emerged as the most commercially promising alternatives, as evidenced by numerous experimental studies demonstrating their ability to enhance energy density and overall performance of LIBs.^{13–17} The carbon component in Si/C composites provides enhanced electrical conductivity and acts as a buffer to accommodate the volume changes of Si during lithiation.^{18–20} However, the Li diffusion behavior in Si/C composites is not very clear, which is of great significance for the anode material design and improvement.

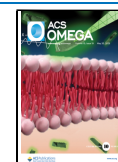
Many studies have made efforts to investigate the properties of Li insertion and diffusion in Si/C composites, attempting to reveal the mechanisms.^{2,21–27} Si et al.²¹ investigated the Li insertion behaviors in graphene/Si composite anodes using first-principles calculations. The study focused on understanding the energetics of Li intercalation, the interaction between Li and the host material, and the diffusion properties

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of Li in graphene/Si composites. Lu et al.²² conducted first-principles calculations to investigate the diffusion of Li in Si encapsulated with graphene. The stable structures of Li adsorbed in graphene/Si slabs and the energy barriers for Li diffusion were studied in order to understand how the presence of graphene affected the diffusion behavior of Li in Si anode material. The Li diffusion mechanism in Si/C composite anodes with the mixture mode and the core–shell mode was investigated by Xu et al.,²³ and the effect of C-layer thickness on Li diffusion behaviors was mainly discussed. Research results showed that the Li diffusion speed in Si/C composites highly depended on the C-layer structure, in which ion channels were needed for Li diffusion. Therefore, compositing Si with a graphene (Gr) material with defects should facilitate the diffusion of Li into the Si/Gr composites. There are usually two typical point defects for Gr, such as single vacancy (SV) and divacancy (DV) defects. SV defects are created by removing a C atom, while DV defects are formed by removing a C–C dimer at first, then annealing to obtain a reconstructed structure.²²

In this work, Li diffusion properties in Si/Gr composites are studied by *ab initio* molecular dynamics (AIMD) in the framework of density functional theory (DFT). Both defective graphene (*d*-Gr) and perfect graphene (*p*-Gr) are adopted in order to investigate the influence of the Gr structure on Li diffusion. First, the structural evolutions of the Si/*p*-Gr composite and Si/*d*-Gr composite during the Li diffusion process are investigated. The effects of various Gr defects on Li diffusion in Si/*d*-Gr composites are presented, and the role played by Gr in Li diffusion is revealed. Moreover, the mean squared displacements (MSDs) of Li atoms are studied, and the Li diffusion coefficients are calculated. The fundamental findings of this work will contribute to guiding the design and development of next-generation anode materials.

2. COMPUTATIONAL METHOD

The Li diffusion properties in Si/Gr composites were studied using the MedeA Vienna *ab initio* simulation package (VASP).²⁸ The interaction between core and valence electrons was described by projector augmented wave (PAW) pseudopotentials.²⁹ The generalized gradient approximation (GGA), based on the Perdew–Burke–Ernzerhof (PBE) functional, was applied for the electron exchange–correlation function.³⁰ A van der Waals-enabled potential was adopted to consider the van der Waals forces. An energy cutoff of 400 eV was used for the expansion of the plane waves.

In order to simulate Li diffusion in Si/Gr composites, various Si/Gr composites were modeled first. Monolayer *p*-Gr and *d*-Gr were placed above an optimized Si(111) surface to model the Si/Gr composite anodes in a mixture mode,²³ as displayed in Figure 1. Two common defects for *d*-Gr, including the SV and DV5-8-5 defects, were considered,²² and *p*-Gr was also modeled as a reference. Here, we abbreviate the above Si/Gr composites with different Gr layers as Si/*p*-Gr, Si/SV, and Si/DV5-8-5, respectively. The thickness of the Si(111) substrate is 10 layers. The length and width of the unit cell for Si/*p*-Gr, Si/SV, and Si/DV5-8-5 are all 6.59×6.59 Å. To accommodate the Gr layer and minimize the lattice mismatch at the Si/Gr interface, a 3% adjustment of the Si bond lengths was employed. The initial Gr and Si interfacial gap was set to 1.5 Å.

The Si/*p*-Gr and Si/*d*-Gr composites used in the following AIMD studies were first optimized using the conjugate

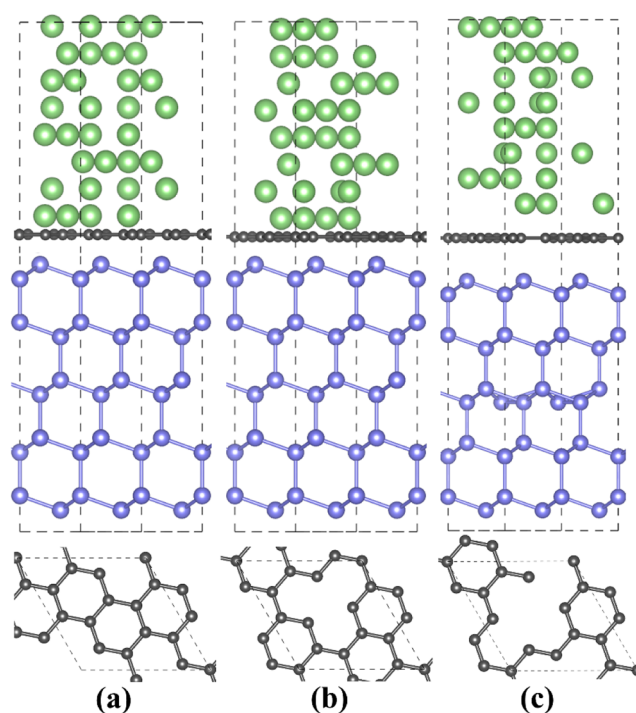


Figure 1. Atomic structures of Si/*p*-Gr (a), Si/SV (b), and Si/DV5-8-5 (c). The bottom figures are the top views of *p*-Gr, SV defect, and DV5-8-5 defect. The gray violet, green, and dark gray spheres represent the Si, Li, and C atoms, respectively.

gradient method, until the residual force was less than 0.02 eV/Å and the energy change per atom was smaller than 10^{-5} eV. The integration of the Brillouin zone was performed with a Gamma-centered *k*-grid mesh of $3 \times 3 \times 1$. During the optimization process, a vacuum layer of 14 Å was applied in the direction perpendicular to the Gr surface. The vacuum space above the Gr layer was then filled with Li atoms to simulate Li diffusion. The numbers of Si, C, and Li atoms (N_{Si} , N_{C} , and N_{Li}) for Si/*p*-Gr, Si/SV, and Si/DV5-8-5 systems are provided in Table 1.

Table 1. Length and Width of the Unit Cell ($L \times W$), and the Numbers of Si, C, and Li Atoms (N_{Si} , N_{C} , and N_{Li}) for Si/*p*-Gr, Si/SV, and Si/dv5-8-5 Composite Systems

	$L \times W$	N_{Si}	N_{C}	N_{Li}
Si/ <i>p</i> -Gr	6.59×6.59 Å	30	14	30
Si/SV	6.59×6.59 Å	30	13	30
Si/DV5-8-5	6.59×6.59 Å	30	12	30

The atomistic structures in Figure 1 and the subsequent figures in our manuscript were produced using the VESTA software.³¹ The Li diffusion in Si/Gr composites was investigated by AIMD simulations with the canonical ensemble (NVT) within the framework of DFT at a temperature of 1200 K. The time step used was 3 fs, and the temperature of the system was controlled by a Nosé–Hoover thermostat. During the simulation process, the bottom Si atoms were set to be free during the AIMD calculations as defined by the mixture mode,²³ and periodic boundary conditions were applied in all three directions. These periodic boundary conditions allowed Li atoms to diffuse from both the upper surface of the Gr layer and the lower surface of the Si lattice. AIMD simulations of 30

ps were carried out for Si/*p*-Gr, Si/SV, and Si/DV5-8-5 composite systems, which were found to be sufficient for Si, C, and Li atoms to mix together.

3. RESULTS AND DISCUSSION

The structural evolution of Li diffusion in various Si/Gr composites at 1200 K during the diffusion process was examined, and the structural snapshots at various simulation times are shown in Figures 2–4. In our simulation, the bottom

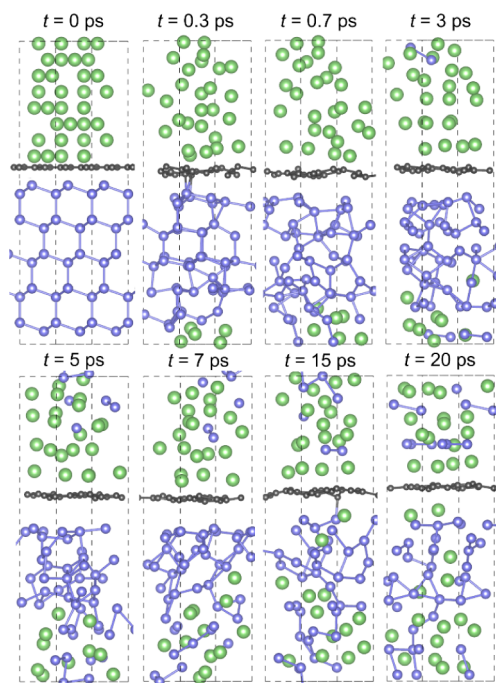


Figure 2. Atomic structures of Li diffusion in the Si/*p*-Gr composite system at various simulation times (*t*) during the diffusion process. In the figure, the gray violet, green, and dark gray spheres represent the Si, Li, and C atoms, respectively.

atoms were set free; hence, under periodic boundary conditions, the Li atoms could diffuse into Si from both the top of the Gr layer and the bottom of the Si lattice. However, for Si/*p*-Gr (Figure 2), the Li atoms tend to diffuse into Si only from the bottom rather than passing through the *p*-Gr layer. These results are consistent with the literature, which demonstrates that it is very difficult for Li to diffuse into the *p*-Gr layer when Si particles are encapsulated with *p*-Gr.²² For Si/SV and Si/DV5-8-5 (Figures 3 and 4), Li atoms could diffuse into Si from both the top of the defective Gr layer and the bottom of the Si lattice. This is because Li diffusion at the Si/Gr interface depends on the atomic structure of the Gr layer. For the *p*-Gr layer, it could not provide ion channels for Li atoms to diffuse in ref 23, while for Si/SV and Si/DV5-8-5, the *d*-Gr layer could provide voids for Li atoms to diffuse in. Therefore, Li atoms could diffuse into Si from both the top and the bottom for Si/SV and Si/DV5-8-5 systems, but for the Si/*p*-Gr system, Li atoms could diffuse into Si only from the bottom of the Si lattice. Moreover, the DV5-8-5 layer could provide more voids compared with the SV layer; therefore, the Li diffusion speed for the Si/DV5-8-5 system is faster. Overall, for both Si/SV and Si/DV5-8-5 systems, more Li atoms tend to diffuse into Si from the bottom rather than passing through the *d*-Gr layer. Therefore, the diffusion speed of Li is closely

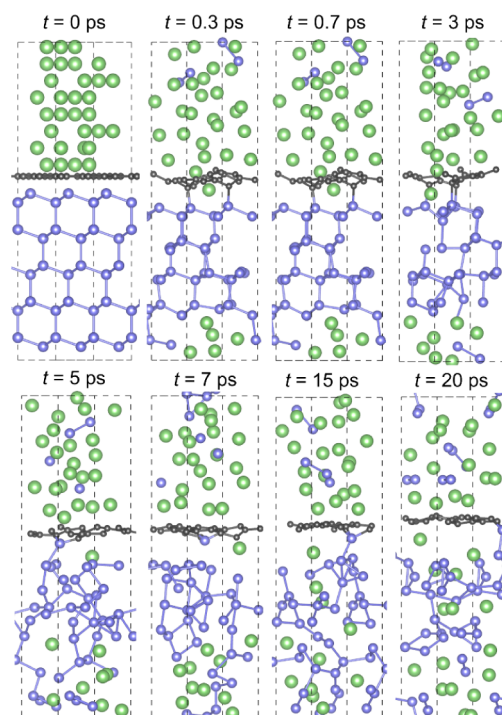


Figure 3. Atomic structures of Li diffusion in a Si/SV composite system at various simulation times (*t*) during the diffusion process. The gray violet, green, and dark gray spheres represent the Si, Li, and C atoms, respectively.

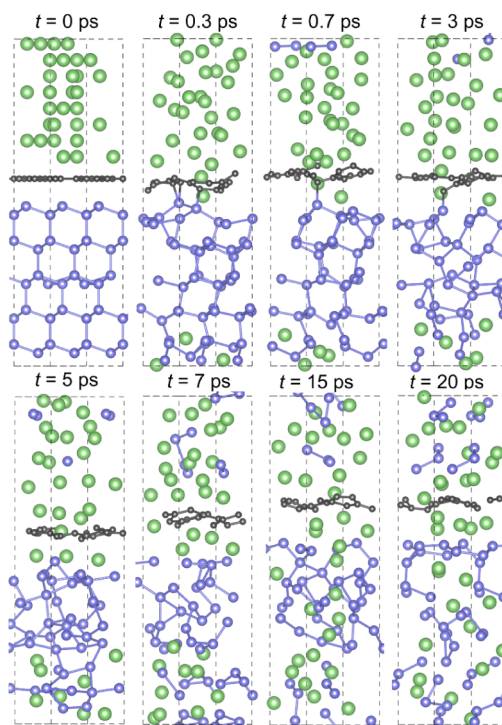


Figure 4. Atomic structures of Li diffusion in the Si/DV5-8-5 composite system at various simulation times (*t*) during the diffusion process. The gray violet, green, and dark gray spheres represent the Si, Li, and C atoms, respectively.

related to the structure of the Gr layer. Compared with Si/*p*-Gr composites, the thorough mixing of Li, C, and Si atoms in Si/*d*-Gr composites occurs faster. The Li, C, and Si atoms could

be completely mixed at around 15 ps, and 30 ps is sufficient to complete the Li diffusion process calculations in the Si/Gr composite materials.

Figures 2–4 all demonstrate that the insertion of Li atoms into Si leads to crystalline Si transforming into an amorphous state. As the Li atoms gradually embed into Si, more and more Si–Si bonds are deformed and broken, allowing the Si system to accommodate the embedded Li. The tetrahedrally bonded Si network disintegrates into low-connectivity Si microstructures, e.g., isolated atoms, dumbbells, boomerangs, stars, and chains, as shown in Figure 5. The structures presented

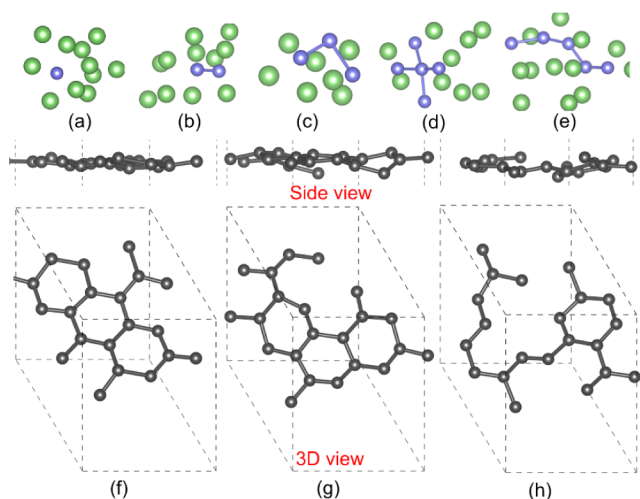


Figure 5. Representative microstructures formed after Li diffusing into Si, including isolated atoms (a), dumbbells (b), boomerangs (c), stars (d), and chains (e); side and 3D views of various Gr layer for Si/*p*-Gr (f), Si/SV (g), and Si/DV5-8-5 (h) at simulation time of 0.7 ps.

above are extracted from Figures 2–4. Similar results were also found in refs 32 and 33. These studies suggest that the generation of these microstructures is accompanied by the

volume expansion and softening of the Si system.^{32,33} During Li diffusion, all Gr layers no longer maintain a planar configuration, as shown in Figures 2–4. In order to better understand the structure of the Gr layer during the Li diffusion process, side and 3D views of various Gr layers for Si/*p*-Gr, Si/SV, and Si/DV5-8-5 at a simulation time of 0.7 ps are shown in Figure 5f–h. The Gr layer structure in Si/SV and Si/DV5-8-5 composite systems changes relatively larger during the Li diffusion process, especially for the DV5-8-5 layer. The C atoms of the Gr layer near the defects deviate from their original positions, resulting in obvious ripples in the Gr layer.

In order to further analyze the structural evolution during the diffusion process, the radial distribution function (RDF, $g(r)$) as a function of simulation time was studied. Figure 6 presents $g(r)$ at 1200 K for Si–Si, Si–Li, and Li–Li pairs in the Si/*p*-Gr, Si/SV, and Si/DV5-8-5 systems at various stages of diffusion. The $g(r)$ plots for Si–Si pairs in the initial structures of all Si/Gr composites show crystalline characteristics with sharp peaks. The other $g(r)$ peaks, except for the first peak for Si–Si pairs, gradually disappear after $t \sim 3$ ps, indicating a transformation of the crystalline phase into an amorphous phase. The $g(r)$ peak values for Si–Si and Li–Li pairs decrease with the diffusion of Li atoms, while those for Si–Li pairs increase over time, showing that Li atoms gradually mix with Si atoms. The newly formed LiSi system shows an amorphous state. The first $g(r)$ peaks for Si–Si pairs shift slightly from 2.35 to 2.5 Å, indicating an increase in the Si–Si bond distance. This occurs because charge is transferred from Li to Si during lithiation, causing Si to be negatively charged and develop a repulsive force.³⁴ The positions of the $g(r)$ peaks for Si–Li pairs are around 2.7–2.8 Å.

If we examine Figure 6, many differences among the $g(r)$ curves of the Si/*p*-Gr, Si/SV, and Si/DV5-8-5 composites can be observed. The $g(r)$ peak value changes over diffusion time for Si–Si pairs and Si–Li pairs are more significant for Si/SV and Si/DV5-8-5 composites compared with the Si/*p*-Gr composite. This observation is consistent with the previous structural snapshot analysis. For Si/SV and Si/DV5-8-5, Li

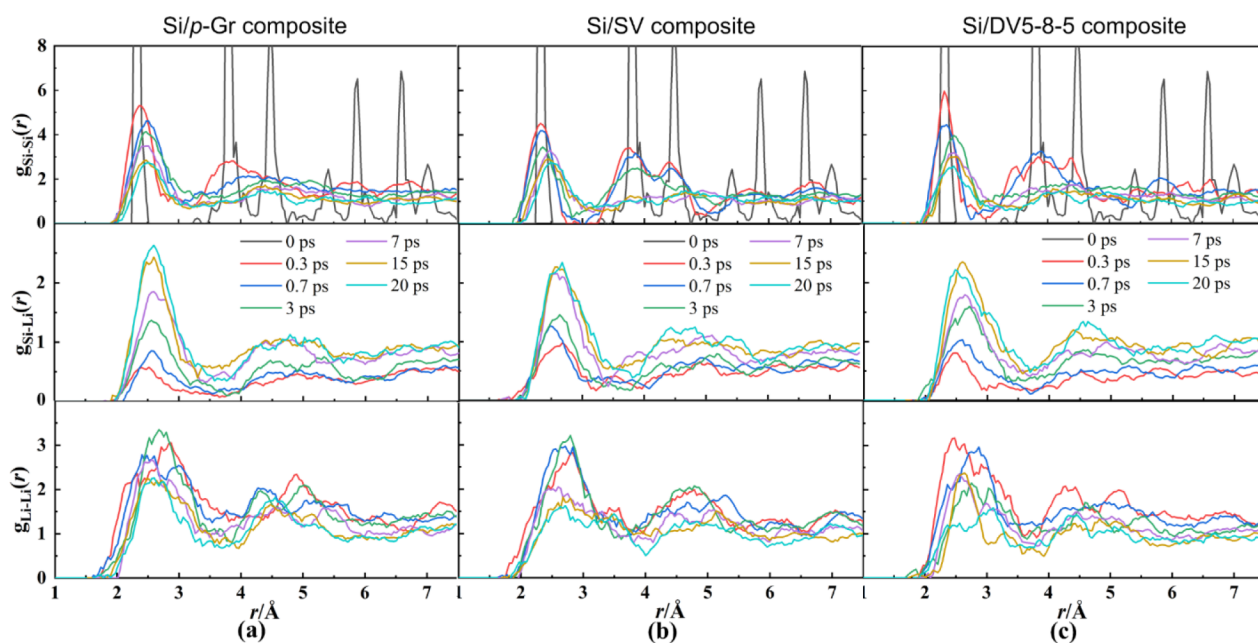


Figure 6. $g(r)$ at 1200 K for Si–Si, Si–Li, and Li–Li pairs in Si/*p*-Gr (a), Si/SV (b), and Si/DV5-8-5 (c) anodes at various stages of diffusion.

atoms could diffuse into Si from both the top of the defective Gr layer and the bottom of the Si lattice. Therefore, the mixing speed of Li and Si atoms is faster, accompanied by a rapid decrease in the $g(r)$ peak value for Si–Si pairs and an increase for Si–Li pairs.

In order to elucidate the effect of the Gr layer structure on Li diffusion in Si, the $g(r)$ curves for C–Li pairs in fully lithiated Si/*p*-Gr, Si/SV, and Si/DV5-8-5 systems at 1200 K are given in Figure 7. The positions of the $g(r)$ peak for C–Li are around

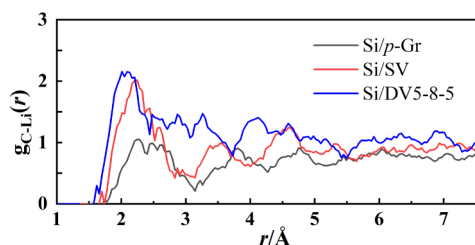


Figure 7. $g(r)$ at 1200 K for C–Li pairs in Si/*p*-Gr, Si/SV, and Si/DV5-8-5 anodes at the full lithiated state.

2.25 Å, which is consistent with the literature.²³ A higher peak value in Figure 7 means that more C–Li bonds are formed in the composite; that is, more C–Li bonds are formed in Si/SV and Si/DV5-8-5 anodes than Si/*p*-Gr. This is because there is no void in the *p*-Gr layer, hence providing no storage space for the Li atoms, while the structural defects of SV and DV5-8-5 layers provide storage voids for Li atoms.

Next, we evaluated the coordination number (CN) of Si atoms in order to further understand the mixing mechanism of Li diffusion into Si/Gr composites. Figure 8 shows the distribution of Si–Si CN with a cutoff radius of 2.6 Å (the Si–Si bond length is around 2.5 Å³³) in Si/*p*-Gr, Si/SV, and Si/DV5-8-5 anodes at various stages of diffusion. The CN in Figure 8 represents the number of Si neighbors. Before Li mixing, the majority of CN for Si atoms is 4, and a small portion has a CN of 3. It is clear from Figure 8 that the fraction of 4-coordinated Si atoms decreases with simulation time. For Si/*p*-Gr composite, 1-, 2-, 5-, and 6-coordinated Si atoms appear in the first 0.7 ps of simulation, and this result is consistent with the literature.³⁵ With the increase of simulation time, the fraction of 1- and 2-coordinated Si atoms continuously increases, while those of 5- and 6-coordinated Si atoms decrease initially and then disappear by the 20 ps mark of the simulation. For Si/SV and Si/DV5-8-5 composites, there are no 6-coordinated Si atoms, and the fraction of 5-coordinated Si atoms is less compared to the Si/*p*-Gr composite. According to previous studies,³⁶ the emergence of 5- and 6-coordinated Si atoms is accompanied by a phase transition under relatively higher pressure, between 8 GPa and about 30 GPa. For Si/*p*-Gr composite, Li atoms tend to diffuse into Si only from the bottom, causing relatively high pressure on the silicon system, which leads to the emergence of 5- and 6-coordinated Si atoms. For Si/SV and Si/DV5-8-5 composites, Li atoms could diffuse into Si from both the top of the defective Gr layer and the bottom of the Si lattice. Therefore, the resulting pressure is lower, and the 5- and 6-coordinated Si atoms are fewer. The reason 5- and 6-coordinated Si atoms disappear and are replaced by low-coordination atoms is that the Si–Si bonds of the Si network are broken by the embedded Li atoms, leading to the formation of low-coordination atoms.

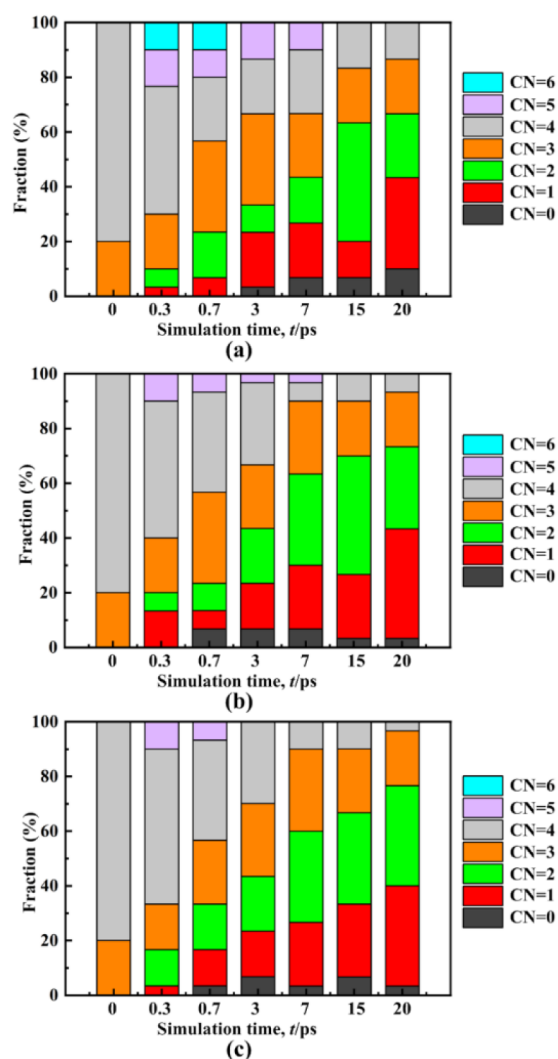


Figure 8. Distribution of Si–Si CN with a cutoff radius of 2.6 Å in (a) Si/*p*-Gr, (b) Si/SV, and (c) Si/DV5-8-5 anodes at various stages of diffusion. Here 0, 1, 2, 3, 4, 5, and 6 are referred to as Si atoms connected to zero, one, two, three, four, five, and six Si atoms, respectively.

Figure 9 shows the average Si–Li CN with a cutoff radius of 2.9 Å³³ in Si/*p*-Gr, Si/SV, and Si/DV5-8-5 anodes at various stages of diffusion. For the Si/*p*-Gr composite, the average Si–Li CN increases from nearly 0 at 0 ps to 4 at 20 ps. For Si/SV and Si/DV5-8-5 composites, the growth rate of the average

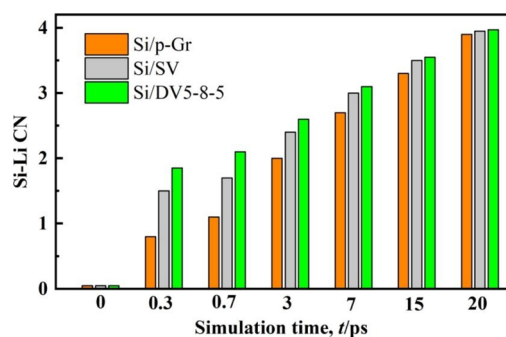


Figure 9. Average Si–Li CN with a cutoff radius of 2.9 Å in Si/*p*-Gr, Si/SV, and Si/DV5-8-5 anodes at various stages of diffusion.

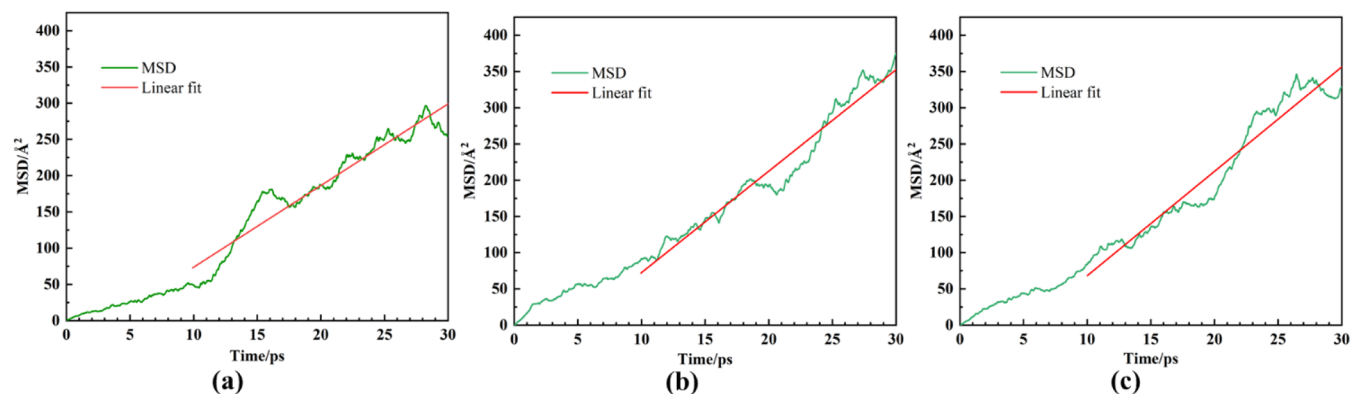


Figure 10. Total MSDs profiles for Li atoms in (a) Si/p-Gr, (b) Si/SV, and (c) Si/DV5-8-5 composite systems at 1200 K.

Si–Li CN is faster compared with that for the Si/p-Gr composite. It is evident that the mixing speed of Si and Li atoms is faster in Si/SV and Si/DV5-8-5 composites with a defective Gr layer, as compared to the Si/p-Gr composite with a perfect Gr layer.

In order to examine the diffusivity of Li in Si/Gr composites, the mean squared displacements (MSDs) of Li atoms as a function of AIMD time steps are calculated using the following expression

$$\text{MSD} = \langle |r_i(t) - r_i(0)|^2 \rangle = \frac{1}{N} \sum_{i=1}^N |r_i(t) - r_i(0)|^2 \quad (1)$$

Where $r_i(t)$ represents the positions of the specific atom i at time t , N is the total atomic number, and the operator $\langle \rangle$ denotes the average value over all the atoms. Then, the diffusivity can be calculated from the MSD using the Einstein relation as follows

$$D = \lim_{t \rightarrow \infty} \frac{\text{MSD}}{nt} \quad (2)$$

Where n is a constant, with $n = 6$ for our three-dimensional Li diffusion study.

Figure 10 shows the total MSD profiles for Li atoms in Si/p-Gr, Si/SV, and Si/DV5-8-5 composite systems at 1200 K. The diffusivity can be obtained by linearly fitting the MSD curves within the time range in which the Si and Li atoms are fully mixed. In our study, the time range of $t > 15$ ps is selected to compute the diffusion coefficient (D_{Li}) for Li atoms in Si/p-Gr, Si/SV, and Si/DV5-8-5 composite systems at 1200 K, as summarized in Table 2. The diffusion coefficient for Li atoms

Table 2. Diffusion Coefficients (D_{Li} , cm^2/s) for Li Atoms in Si/p-Gr, Si/SV, and Si/dv5-8-5 Composite Systems at 1200 K

	Si/p-Gr	Si/SV	Si/DV5-8-5
D_{Li} (cm^2/s)	1.88×10^{-4}	2.32×10^{-4}	2.47×10^{-4}

in Si/p-Gr is about $1.88 \times 10^{-4} \text{ cm}^2/\text{s}$ at 1200 K, which is close to the calculated value for Si/C composite reported in the literature (about $2.097 \times 10^{-4} \text{ cm}^2/\text{s}$).²³

The diffusion coefficient of Li-ions is relatively low at room temperature, potentially requiring extremely long AIMD simulation times to observe significant diffusion behavior. Since the computational cost of AIMD simulations is very high, AIMD studies on Li diffusion are generally conducted at

high temperatures. Within the limitations of AIMD simulations (e.g., the small cell size and simulation time), we cannot expect to obtain precise values of the diffusion coefficients.³⁷ However, the underlying physics influencing the Li diffusion process can be observed.

While the diffusion coefficients for Li atoms in Si/SV and Si/DV5-8-5 composites at 1200 K are 2.32×10^{-4} and $2.47 \times 10^{-4} \text{ cm}^2/\text{s}$, respectively, these results indicate that the introduction of defects into Si/Gr composites improves Li diffusion in the Si material by up to 32%. Gupta and his coworkers also showed that vacancies and defects can significantly enhance diffusion.^{37–39} The MSD slope of the vacant structure is approximately four times greater than that of the perfect crystalline structure,³⁷ because Li diffusion in the perfectly crystalline phase is limited, as it has to overcome the interaction energy of nearby host elements. In contrast, structures with vacancies and defects could provide ion channels for Li diffusion. Compared with SV defects, DV defects have a stronger effect on improving the diffusivity of Li atoms. This is because the Gr layer with DV defects could provide more efficient channels through which Li atoms can pass.

4. CONCLUSIONS

In this paper, Li diffusion properties in Si/*d*-Gr composites are investigated by *ab initio* molecular dynamics to illustrate how the defective Gr layer promotes Li diffusion behaviors. Two common defects, including the SV and DV5-8-5 defects, were considered. The results for the structural evolutions of Li diffusion demonstrate that in the Si/p-Gr composite, the Li atoms tend to diffuse into Si only from the bottom. For Si/SV and Si/DV5-8-5, Li atoms could diffuse into Si from both the top of the defective Gr layer and the bottom of the Si lattice, because the *d*-Gr layer could provide voids for Li atoms to diffuse in. The Gr layer structure in Si/SV and Si/DV5-8-5 composite systems changes more significantly during the Li diffusion process compared with that in the Si/p-Gr composite. The RDF results show that the $g(r)$ peak value changes over diffusion time for Si–Si pairs and Si–Li pairs is more significant for Si/SV and Si/DV5-8-5 composites compared with the Si/p-Gr composite, further explaining the analysis of structural evolutions during the diffusion process. In order to further understand the mixing mechanism of Li diffusion into Si/Gr composites, the CNs of Si atoms are also evaluated, and the results show that with the increase of simulation time, the fraction of 1- and 2-coordinated Si atoms continuously increases, while those of 5- and 6-coordinated Si atoms

decrease initially and then disappear within 20 ps of simulation. Finally, the MSD and diffusion coefficients are calculated, revealing that the defective Gr layer improves Li diffusion in Si material by up to 32%. Compared with SV defects, DV defects have a stronger effect on improving the diffusivity of Li atoms.

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Notes

The authors declare no competing financial interest.

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